

3.5 Bachelor of Science Degree in Electrical and Computer Engineering

3.5.1 Introduction

This curriculum is developed to cater the demand of innovative, highly skilled, practice oriented, entrepreneur, and ethical manpower in the various fields of Electrical and Computer Engineering in five streams, namely, Communications Engineering, Computer Engineering, Control Engineering, Electronics Engineering, and Power Engineering.

Students will have options to choose their stream/focus area from the above five fields of specializations after successfully completing their fourth year second semester. A successful graduate of the program will not only be globally competent but also have active and decisive role in local industries. It will promote industrialization by motivating staff and students and offering consultancy services for prospective investors and entrepreneurs in the stage of industrial setup.

Moreover, this curriculum has special place in the history of the University in that it is designed according to the Washington Accord as an effort to accredit the entire Engineering program running in the University.

3.5.2 Program Education Objective (PEO)

The program education objectives shall describe accomplishments that the five years program graduates are expected to achieve in the first 3 to 5 years after graduation. The Department of Electrical and Computer Engineering has set the following program education objectives:

PEO1: The graduates will become professional engineers.

PEO2: The graduates will establish their own start-up companies.

PEO3: The graduates will be employed in high-ranking companies and universities locally and internationally.

PEO4: The graduates will be involved in research, design, and development works.

3.5.3 Program Outcome (PO)

Program outcomes are statements that express the knowledge, skills and attitudes of graduates of this program. Apparently, they are extensions and summaries of graduate profiles presented above. For this program twelve Program Outcomes are identified as presented below:

PO1: Apply fundamental knowledge of mathematics, physics and various computational methods to analyze and solve electrical and computer engineering related problems.

PO2: Identify, formulate, research literature and analyze complex electrical and computer engineering problems reaching substantiated conclusions using first principles of mathematics, applied physics and engineering sciences.

PO3: Design solutions for complex engineering problems and systems, components and processes that met specified needs, and develop new solutions with appropriate considerations for public, health, safety, cultural, societal and environmental considerations.

PO4: Conduct experiments, analyze and interpret results, and apply research outcomes to develop new technologies and improve existing ones.

PO5: Create, select and apply appropriate techniques, resources and modern engineering and ICT tools, including prediction and modeling, to complex electrical and computer engineering problems, with an understanding of the limitations.

PO6: Apply contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practices and solutions to engineering problems.

PO7: Understand and evaluate the sustainability and impact of professional electrical and computer engineering works while solving complex engineering problems in societal and environmental contexts.

PO8: Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments which must consider the impact of engineering solutions in global economic, environmental, and societal contexts.

PO9: Ability to work independently and as a team, and as a leader, motivating professionalism in multidisciplinary settings.

PO10: Communicate effectively, in both written and orally, on complex electrical and computer engineering activities with a variety of audiences.

PO11: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects in multi-disciplinary environments.

PO12: Recognize the personal, national, and global needs for, and the preparation and ability to engage in, independent and life-long learning in the broadest context of technological dynamism.

3.5.4 Graduation Requirement

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the University Senate Legislation Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out courses.
- A cumulative grade point average CGPA of 2.00 must be obtained overall.
- A cumulative grade point average CGPA of 2.00 in major area courses.
- Pass the exit exam
- No "F" grade in any course/module taken for undergraduate program.
- Student who fails to graduate due to less CGPA than the required or due to "F" grade have no more chance to upgrade their CGPA or to remove their "F". However, they may be given Certificate of Attendance with their transcript stated as "Graduation Failure"
- Successfully defended his/her B.Sc. Thesis.

List of Courses and Category

Table: List of Common Courses

No	Course Code	Course Name	Course Credit
1	Phil1009	Logic and Critical Thinking	3
2	Psyc1011	General Psychology	3
3	FLEn1003	Communicative English Language Skill I	3
4	GeEs1005	Geography of Ethiopia and the Horn	3
5	Math1007	Mathematics for Natural Science	3
6	SpSc1013	Physical Fitness and Conditioning	P/F
7	Phys1001	General Physics	3
8	FLEn1004	Communicative English Language Skill II	3
9	MCiE1012	Moral and Civic Education	2
10	Incl1010	Inclusiveness	2
11	Anth1002	Social Anthropology	2
12	Econ2009	Economics	3
13	GLTr2011	Global Trend	2
14	Hist2002	History of Ethiopia and the Horn	3
Total Credit Hours			35

List of Core Compulsory Courses

Table: List of Core Courses

No	Course Code	Course Name	Course Credit
1	EmTel108	Emerging Technology for Engineers	3
2	Entr1106	Entrepreneurship for Engineers	3
3	Math1014	Applied Mathematics I	4
4	MEng2001	Engineering Drawing	3
5	Comp2003	Introduction to Computer Programming	3
6	CEng2005	Engineering Mechanics I - statics	3

7	MEng2102	Engineering Mechanics-II (Dynamics)	3
8	Math2007	Applied Mathematics II	4
9	Math2042	Applied Mathematics-IIIB	4
10	MEng2114	Engineering Thermodynamics	3
11	ECEg2102	Fundamentals of Electrical Engineering	4
12	ECEg2110	Probability and Random Processes	3
13	ECEg3101	Computational Methods	3
14	ECEg3103	Applied Electronics I	4
15	ECEg3105	Signals and Systems Analysis	3
16	ECEg3107	Electromagnetic Fields	3
17	ECEg3109	Object Oriented Programming	3
18	ECEg3111	Research Methods and Presentations	2
19	ECEg3113	Electrical Workshop Practice I	1
20	ECEg3102	Applied Electronics II	3
21	ECEg3104	Digital Logic Design	4
22	ECEg3106	Network Analysis and Synthesis	3
23	ECEg3108	Digital Signal Processing	4
24	ECEg3110	Electrical Machines I	4
25	ECEg3112	Electrical Workshop Practices II	2
26	ECEg4101	Introduction to Communication Systems	3
27	ECEg4103	Computer Architecture and Organization	3
28	ECEg4105	Introduction to Control Systems	3
29	ECEg4107	Electrical Measurement and Instrumentation	3
30	ECEg4109	Power Systems I	3
31	IETP4115	Integrated Engineering Team Project	3
32	ECEg4102	Microprocessors and Interfacing	4
33	ECEg4112	Integrated Design Project	3
34	ECEg4100	Industry Internship	6
35	IEng5104	Industrial Management and Engineering Economy	3
36	ECEg5108	Final year project II	6

37	ECEg5107	Final year project I	P/F
	Total		119

List of Communications Engineering Focus Courses

Table List of Communications Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg4304	Digital Communications System	4
2	ECEg4308	EM Waves and Guide Structures	4
3	ECEg5311	Telecommunication Networks	4
4	ECEg5301	Microwave Devices and Systems	3
5	ECEg5303	Fiber Optics Communications	4
6	ECEg5305	Antennas and Radio Wave Propagations	4
7	ECEg5307	Wireless and Mobile Communications	4
8	ECEg5302	Switching and Intelligent networks	3
Total			30

List of Computer Engineering Focus Courses

Table List of Computer Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg4406	Data Communications and Computer Networks	4
2	ECEg5410	Advanced Computer Networks	3
3	ECEg4404	Data Structures and Algorithm	4
4	ECEg4410	Data Base Systems	3
5	ECEg5409	Software Engineering	3
6	ECEg5401	Operating Systems	3
7	ECEg5403	Embedded Systems	4
8	ECEg5405	VLSI Design	3
9	ECEg5407	Introduction to Machine Learning	3
10	ECEg5402	New Trends in Computer Engineering	2

11	ECEg5412	Wireless Communications and Mobile Computing	4
Total			36

Control Engineering Courses

Table: List of Control Engineering Focus Engineering

№	Course Code	Course Name	Course Credit
1	ECEg4510	Modern Control Systems	3
2	ECEg4506	Process Control Fundamentals	3
3	ECEg5509	Industrial Automation	4
4	ECEg5507	Digital Control Systems	3
5	ECEg5511	Robotics and Computer Vision	3
6	ECEg5502	Instrumentation Engineering	3
7	ECEg5510	Artificial Intelligence for Control Engineering	3
8	ECEg5503	Embedded Systems for Control Engineering	3
Total			25

Electronics Engineering Courses

Table: List of Electronics Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg5606	Analog Systems Design	3
3	ECEg5609	Optoelectronics	3
4	ECEg5602	Digital System Design	3
5	ECEg5604	IC Technology	3
6	ECEg5605	Microelectronic Devices and Circuits	3
7	EEEg5608	Power Electronics	3
Total			18

Power Engineering Courses

Table: List of Power Engineering Courses

No	Course Code	Course Name	Course Credit
1	ECEg4704	Electrical Machines II	4
2	ECEg5709	Power Systems Automation	4
3	ECEg4708	Power System II	4
4	ECEg5703	Energy Conversion and Rural Electrification	4
5	ECEg5711	Power System Protection	3
6	EEEG5701	Power Electronics and Electric Drives	4
7	ECEg5702	Power Systems Operation and Control	4
8	ECEg5705	Electrical Installation	3
Total			30

Credit Hour Distribution

Table: Credit Hour Distribution of Communication Engineering Focus

Category		Total Cr. Hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.38%
2	Core Elective/focus area course	36	18.51%
3	University requirement (Core)	3 Cr.hr	1.59%
Core Course Total Cr. Hr		154	81.48%
4	National Requirement	35 Cr.hr	18.51%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Computer Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Control Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Electronics Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Power Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.7%
2	Core Elective/focus area course	36	18.09%
3	University requirement (Core)	3 Cr.hr	1.6%
Core Course Total Cr. Hr		153	81.38%
4	National Requirement	35 Cr.hr	18.62%
Total Cr. Hr		190	

3.5.5 Program Curriculum Structure: Bachelor of Science in Electrical and Computer Engineering

Table 16: Semester Course Breakdown

First Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
Phil1009	Logic and Critical Thinking	3	2	3	0
Psyc1011	General Psychology	3	2	3	0
FLEn1003	Communicative English Language Skills I	3	2	3	0
GeES1005	Geography of Ethiopia and the Horn	3	2	3	0
Math1007	Mathematics for Natural Science	3	2	3	0
SpSc1013	Physical Fitness	P/F	0	0	2
Phys1001	General Physics	3	2	0	3
Total		18	12	15	5

First Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
EmT1008	Introduction to Emerging Technology	3	2	3	0
FLEn1004	Communicative English Language Skills-II	3	3	0	0
Math1014	Applied Mathematics IB	4	3	3	0
MCiE1012	Moral and Civic Education	2	2	0	0
Incl1010	Inclusiveness	2	2	0	0
Anth1002	Social Anthropology	2	2	0	0
Entr1006	Entrepreneurship	3	2	3	0
Total		19	16	9	0

Second Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
Comp2003	Introduction to Computer Programming	3	2	0	3
GLTr2011	Global Trend	2	2	0	0
MEng2001	Engineering Drawing	3	2	0	3
CEng2005	Engineering Mechanics I (Statics)	3	2	3	0
Math2007	Applied Mathematics IIB	4	3	3	0
Econ2009	Economics	3	2	3	0
Total		18	13	9	6

Second Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg2102	Fundamentals of Electrical Engineering	4	2	3	3
MEng2102	Engineering Mechanics-II (Dynamics)	3	2	3	0
Math2042	Applied Mathematics IIIB	4	3	3	0
ECEg2110	Probability and Random Processes	3	2	3	0
MEng2114	Engineering Thermodynamics	3	2	3	0
Hist2002	History of Ethiopia and the Horn	3	2	3	0
Total		17	11	12	6

Third Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg3101	Computational Methods	3	2	0	3
ECEg3103	Applied Electronics I	4	2	3	3
ECEg3105	Signals and System Analysis	3	2	3	0
ECEg3107	Electromagnetic Fields	3	2	3	0
ECEg3109	Object Oriented Programming	3	2	0	3
ECEg3111	Research Methods and Presentation	2	2	0	0
ECEg3113	Electrical Workshop Practices I	1	0	0	3
Total		19	12	9	12

Third Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg3102	Applied Electronics II	3	2	0	3
ECEg3104	Digital Logic Design	4	2	3	3
ECEg3106	Network Analysis and Synthesis	3	2	3	0
ECEg3108	Digital Signal Processing	4	2	3	3
ECEg3110	Electrical Machines I	4	2	3	3
ECEg3112	Electrical Workshop Practices II	2	1	0	3
Total		20	11	12	15

Fourth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4101	Introduction to Communication Systems	3	2	0	3
ECEg4103	Computer Architecture and Organization	3	2	3	0
ECEg4105	Introduction to Control Systems	3	2	0	3
ECEg4107	Electrical Measurement and Instrumentation	3	2	0	3
ECEg4109	Power Systems I	3	2	0	3
IETP4115	Integrated Engineering Team Project	3	0	0	9
Total		18	10	3	21

Specialization Courses

Communication Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4304	Digital Communication Systems	3	2	0	3
ECEg4406	Data Communications and Computer Networks	4	2	3	3
ECEg4308	EM waves and Guide Structures	3	2	3	0
ECEg4112	Integrated Design Project	3	0	0	9
Total		17	8	9	18

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5301	Microwave Devices and Systems	3	2	0	3
ECEg5303	Fiber Optics Communications	3	2	0	3
ECEg5305	Antennas and Radio Wave Propagations	4	2	3	3
ECEg5307	Wireless and Mobile Communications	4	3	0	3
ECEg5605	Microelectronic Devices and Circuits	3	2	0	3
ECEg5311	Telecommunication Networks	3	2	3	0
ECEg5107	Final year project I	P/F			
Total		20	12	9	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5302	Switching and Intelligent Networks	3	2	0	3
ECEg5410	Advanced Computer Networks	3	2	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	0	27

Computer Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4404	Data Structures and Algorithm	4	2	3	3
ECEg4410	Database Systems	3	2	0	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4406	Data Communications and Computer Networks	4	2	3	3
Total		18	8	9	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec.	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec.	Tut	Lab
ECEg5409	Software Engineering	3	2	3	0
ECEg5401	Operating Systems	3	2	0	3
ECEg5403	Embedded Systems	4	2	3	3
ECEg5405	VLSI Design	3	2	0	3
ECEg5407	Introduction to Machine Learning	3	2	0	3
ECEg5511	Robotics and Computer Vision	3	2	0	3
ECEg5107	Final year project I	P/F			
Total		19	12	6	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5402	New Trends in Computer Engineering	2	2	0	0
ECEg5412	Wireless Communications and Mobile Computing	4	3	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	10	3	21

Control Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4510	Modern Control Systems	3	2	0	3
ECEg4704	Electrical Machines II	4	2	3	3
ECEg4506	Process Control Fundamentals	3	2	0	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4102	Microprocessors and Interfacing	4	2	3	3
Total		17	8	6	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
EEeg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5701	Power Electronics and Electric Drives	4	2	3	3
ECEg5705	Electrical Installation	3	2	0	3
ECEg5503	Embedded Systems for Control Engineering	3	2	0	3
ECEg5507	Digital Control Systems	3	2	0	3
ECEg5511	Robotics and Computer Vision	3	2	0	3
ECEg5509	Industrial Automation	4	2	3	3
ECEg5107	Final year project I	P/F			
Total		20	12	6	18

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5502	Instrumentation Engineering	3	2	0	3
ECEg5510	Artificial Intelligence for Control Engineering	3	2	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	3	24

Electronics Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4304	Digital Communication Systems	3	2	0	3
ECEg4308	EM Waves and Guide Structures	3	2	3	0
ECEg4112	Integrated Design Project	3	0	0	9
ECEg5606	Analog System Design	3	2	3	0
ECEg5608	Power Electronics	3	2	0	3
Total		19	8	9	18

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5301	Microwave Devices and Systems	3	2	0	3
ECEg5307	Wireless and Mobile Communications	4	3	0	3
ECEg5609	Optoelectronics	3	2	3	0
ECEg5605	Microelectronic Devices and Circuits	3	2	0	3
ECEg5405	VLSI Design	3	2	0	3
ECEg5107	Final year project I	P/F			
Total		16	11	12	9

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5602	Digital Systems Design	3	2	3	0
ECEg5604	IC Technology	3	2	3	0
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	9	18

Power Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4510	Modern Control Systems	3	2	0	3
ECEg4704	Electrical Machines II	4	2	3	3
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4708	Power Systems II	4	2	3	3
Total		18	8	9	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec.	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5703	Energy Conversion and Rural Electrification	4	2	3	3
ECEg5711	Power System Protection	3	2	0	3
ECEg5701	Power Electronics and Electric Drives	4	2	3	3
ECEg5705	Electrical Installation	3	2	0	3
ECEg5709	Power Systems Automation	4	2	3	3
ECEg5107	Final year project I	P/F			
Total		18	10	9	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5502	Instrumentation Engineering	3	2	0	3
ECEg5702	Power Systems Operation and Control	4	2	3	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		13	4	6	21

3.5.6 Program Courses Synopsis

Course Code: Math-1051

Course Name: Applied Mathematics IB

Credit Hour: 4

Prerequisite: None

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: MEng 2101

Course Name: Engineering Drawing

Credit Hour: 3 (1 Lecture, 1 Tutor and 5 Practical)

Prerequisite: None

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view.

Course Code: Comp 2003

Course Name **Introduction to Computer Programming**

Credit Hour: 3

Prerequisite: None

This course, after a brief introduction of what programming languages are and their classifications extends fundamental programming principles and concepts. It specifically deals with variables and constants; scope; operators and operator precedence; comment statements; input and output statements; control structures and arrays; functions and pointers; and structures and files.

Course Code: Meng2104

Course Name Engineering Mechanics-I (Statics)

Credit Hour: 3

Prerequisite: General Physics (Phys-1101)

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements.

Course Code: Comp 2003

Course Name Engineering Mechanics II (Dynamics)

Credit Hour: 3

Prerequisite: CEng2103: Engineering Mechanics I (Statics)

Math1051: Applied Mathematics-I

Basic equations of motion; Kinematics of particles and rigid bodies; Kinetics of particles and rigid bodies

Course Code: Math-2011

Course Name Applied Mathematics II

Credit Hour: 4

Prerequisite: Applied Mathematics I (Math 1051)

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: Math-2022

Course Name Applied Mathematics IIIB

Credit Hour: 4

Prerequisite: Applied Mathematics II (Math-2011))

This course covers basic elements of ordinary differential equations, Laplace and Fourier transforms, vector differential calculus, Line and surface integral and complex analytic functions.

Course Code: MEng2105

Course Name **Engineering Thermodynamics**

Credit Hour: 3

Prerequisite Applied Mathematics I (Math-1051)

Fundamental concepts; Pure substances; Vapor pressure curves; Steam tables; Phase diagrams of steam; First law of Thermodynamics: closed and open systems, enthalpy; Second law of Thermodynamics: Reversible and irreversible processes; Carnot cycle; Entropy; Ideal gases and their mixtures; gas-steam mixtures; wet air; psychometric charts and air conditioning process; Vapor power and refrigeration cycles; Air standard cycles;

Course Code: Course Code: ECEg2102

Course Name **Fundamentals of Electrical Engineering**

Credit Hour: 4

Prerequisite Math-1014: Applied Mathematics I for Engineering

In this course we discuss about: Review of Electromagnetic Phenomenon and Variables, Electric Circuit parameters, Circuit Analysis, SteadyState Single Phase AC Circuit Analysis, Transient Circuit Analysis, Introduction to poly phase systems.

Course Code: ECEg2110

Course Name Probability and Random Process

Credit Hour: 3

Prerequisite

Course Code: ECEg3101

Course Name Computational Methods

Credit Hour: 4

Prerequisite Applied Engineering Mathematics III (Math-2022) and C++ Programming (ECEg2108)

The course aims at introducing students in finding numerical solutions to problems for which analytical solutions either do not exist or are not readily or cheaply obtainable. It enables students to apply linear algebra and calculus. It also aims to help student develop programming skills.

Course Code: ECEg3103

Course Name Applied Electronics I

Credit Hour: 4(L 2,P 3,T 3)

Prerequisite Fundamentals of Electrical Engineering (ECEg2102)

The course focuses on discussing basic semiconductor theory, semiconductor diodes and their applications, BJT, FET, Amplifier frequency response, Multistage amplifiers, Power amplifier and Tuned amplifiers.

It also introduces the basic measurements on characteristics of electronic devices and circuits, semiconductor diode characteristics, rectification and filtering, bipolar junction transistor characteristics, transistor biasing and operating point stability, transistor amplifier, frequency response, multistage amplifier, power amplifier and Tuned amplifier.

Course Code: ECEg3105

Course Name **Signals and Systems Analysis**

Credit Hour: 3

Prerequisite Math-2022: Applied Mathematics III

The course aims and enables student to understand and apply the representation, classification, characterization and analysis of signals and systems in time and frequency domains.

Course Code: ECEg3107

Course Name **Electromagnetic Fields**

Credit Hour: 3

Prerequisite ECEg2102: Fundamentals of Electrical Engineering

This course lays the foundation for understanding basic electromagnetic concepts, fundamental governing laws and guiding principles which students can apply in subsequent courses. The course covers: Vector Algebra, Scalar and Vector Fields, Coordinate Systems and Transformation, Vector Calculus, Line, Surface and Volume Integrals, Vector Differential Operator, Electrostatic fields, Coulomb's and Gauss's Laws for static electric charge distributions in vacuum and material bodies, Electrostatic Boundary Value Problems, Poisson's and Laplace's equations, Resistance and Capacitance, Magneto static Fields, Biot-Savart's and Ampere's Laws for steady state current source in free and material space, Magnetic Forces, Materials, and Devices, Faraday's Law, Maxwell's Equations for Static and dynamic (time-varying) Electromagnetic Fields

Course Code: ECEg3109

Course Name **Object Oriented Programming**

Credit Hour: 3

Prerequisite C++ Programming (ECEg2108)

This course is designed to provide the fundamental theories and principles and techniques of object-oriented programming paradigms. Topics to be dealt with are: classes, objects, data abstraction, encapsulation and information-hiding, overloading, subclasses and inheritance, polymorphism, class hierarchies and exception handling. This course also gives chances to students to work on object-oriented programming language during practical session to implement; test and experiment an object oriented paradigm of the program design and implementation.

Course Code: ECEg4111

Course Name **Research Methods and Presentation**

Credit Hour: 2

Prerequisite None

This course deals with discussion of research and development methods. Moreover, it addresses current and emerging research topics in the area of computer engineering. Selection of topics for individual projects depends on the instructors and department. Literature review and analysis, intensive readings, discussions and reports will be the main activities and assessment methods.

Course Code: ECEg3112

Course Name **Electrical Workshop Practice I**

Credit Hour: 1

Prerequisite None

In this course we discuss about: General Safety rules, Common types of Electrical hand Tools, Wiring materials and accessories, Termination, splicing of solid and flexible wires, Soldering; Residential installation which includes Lighting, Power and Signal circuits; Fire Alarm installation, Security camera installation, testing and inspecting these electrical installations.

Course Code: ECEg3102

Course Name **Applied Electronics II**

Credit Hour: 3

Prerequisite ECEg3103-Applied Electronics I

The course focuses on discussing advanced electronics circuits and concepts of feedback system. Discusses the working principles of differential and operational amplifiers. Design and simulate the circuits of differential and operational amplifiers using Multisim. Discuss, design and simulate applications of amplifiers.

Introduces to the wave shaping circuits such as oscillators and Multi vibrators, timers and switching circuits.

Course Code: ECEg3104

Course Name **Digital Logic Design**

Credit Hour: 4

Prerequisite Applied Electronics II (ECEg3102)

Digital Logic Design is a comprehensive study of the principles and techniques of modern digital system. It provides an overview of the principles underlying number systems, arithmetic operations, decimal and alphanumeric codes, Boolean algebra, combinational and sequential circuits. Furthermore, analysis and design of combinational sequential logic system is discussed.

Course Code: ECEg3106

Course Name **Network Analysis and Synthesis**

Credit Hour: 3

Prerequisite ECEg3105: Signals and Systems Analysis

The course covers the techniques of modeling, analysis, design and synthesis of N-Port passive and active and passive electrical filters in a classical and a modern approach

Course Code: ECEg3108

Course Name **Digital Signal Processing**

Credit Hour: 4

Prerequisite ECEg3105: Signals and Systems Analysis

The course mainly deals with digital signal processing: sampling and quantization, A/D and D/A converters, Discrete-time systems convolution, Z-transforms and its Implementation, Digital filter realizations Fast Fourier transforms digital filter design, decimation and interpolation, random signals, and some applications. It also introduce FIR and IIR Filters, Filter Design Methods, Interpolation and Decimation, Adaptive Signal Processing and its application, Image and Speech Processing.

Course Code: ECEg4113

Course Name **Electrical Machines I**

Credit Hour: 4

Prerequisite ECEg3107: Electromagnetic Fields

In this course we discuss about: Principle of magnetics, Transformers, Induction Machine, DC Machine and synchronous Machine

Course Code: ECEg4107

Course Name **Electrical Workshop Practice II**

Credit Hour: 2

Co. requisite ECEg4113 Electrical Machine I

In this course we discuss about: Rewinding of transformers and Stator of three phase Squirrel cage Induction motor; starting and control of 3-phase Induction Motor, Explain and perform Computer maintenance and Repair of electrical apparatus coil (Relay coil)

Course Code: ECEg4101

Course Name **Introduction to Communication Systems**

Credit Hour: 3

Pre requisite ECEg3102: Applied Electronics II

ECEg3105: Signals and System Analysis

ECEg2110: Probability and Random Processes

This course is intended to give a strong background on Communication Systems Engineering. It starts by

introducing basic concepts of communication systems, and it contains in-depth study of analog communication systems such as: AM, PM, and FM modulation/demodulation techniques. It also introduces digital modulation techniques and data communication.

Course Code: ECEg4103

Course Name **Computer Architecture and Organization**

Credit Hour: 3

Pre requisite Digital logic design (ECEg3104)

This course introduces students to the organization and architecture of computer systems, beginning with the standard von Neumann model and then moving forward to more recent architectural concepts. **General Introduction:** Organization and architecture, computer evolution, measuring performance, models of a computer system; **The Central Processing Unit:** Computer arithmetic, Instruction sets, Instruction format and addressing modes, RISC and CISC, Pipelining, The Control Unit (Hardwired and Micro-Programd Implementations); assembly/machine language programming; **Memory Systems:** Classification and hierarchy of Memory systems, Main memory, Cache Memory, Secondary Memory, Other types of memory, Memory Management I/O and interrupts; **Input Output Systems:** Input Output devices, modes of transfer, I/O interface, Techniques used for I/O Operations: Programd, Interrupt-driven, Direct Memory Access.

Course Code: ECEg4153

Course Name **Introduction to Control Systems**

Credit Hour: 3

Pre requisite Computational Method, Applied Electronics I

This course mainly provides students with the knowledge and skills required to the basic concept of control systems, mathematical modeling of physical system, characteristics of feedback and non-feedback system, time and frequency response analysis of control system. At last the students will able to design and analyze the stability of linear control systems.

Course Code: ECEg3110

Course Name **Electrical Measurement and Instrumentation**

Credit Hour: 3

Pre requisite Fundamentals of Electrical Engineering (ECEg2102)

This course mainly provides students with the basic operational principles and limitations of electrical and electronic instruments, sensors and transducers with their application, signal conditioning and conversion, output presentation, and instrument transformer.

Course Code: ECEg4109

Course Name **Power Systems I**

Credit Hour: 3

Pre requisite Electrical Machine I (ECEg4113)

This course introduces and explains fundamentals of electrical power systems design and engineering. Phasors and their application to power systems analysis are reviewed. The concept of the per-unit system is introduced and applied to circuit calculations. Transmission line parameters, their calculation, and transmission line modeling are introduced. Steady-state operation of transmission lines is modeled and investigated. Mechanical design aspect of overhead transmission line discussed. Brief introduction of underground cable construction and calculation of its insulation resistance is explained.

Course Code: ECEg4102

Course Name **Microprocessors and Interfacing**

Credit Hour: 4

Pre requisite Computer architecture and Organization (ECEg4103)

This course is designed to impart in-depth knowledge in the design, programming and organization of microcomputers and interfacing circuits. This course completely covers the popular Intel μ P 8086, which would be a stepping-stone for learning the X 86 families of microprocessors. Fundamentals of Microprocessors and Microcontrollers Architecture; Register level organization; Memory organization; Bus systems; Peripheral hardware organization; Instruction Set; Assembly language programming; C programming; Microcontroller programming starting from the architecture to all its peripherals; Different applications of microcontroller in real world, lower level communication protocols for microcontrollers; Interfacing Fundamentals; Programmable Interrupt Controller (PIC); Programmable Peripheral Interface (PPI).

Course Code: ECEg4112

Course Name **Integrated Design Project**

Credit Hour: 3

Pre requisite Senior standing

This Integrated Design Project is either a group-based (up to 3) or individualized engineering design project to evaluate the overall knowledge of the student(s) before going for the BSc Thesis. If it is an individualized project the student will be evaluated on the basis of his/her understanding, application and presentation of basic computing knowledge and skills. If it is a group-based project each student will also be evaluated its role in the group work and its team work dimensions.

Course Code: ECEg4100

Course Name **Industry Internship**

Credit Hour: 6

Pre requisite Senior standing

The Students must be attached with an industry for at least five months in order to get real world experience in their field of study, which compliments their education. The student and the internship coach at the industry have to submit a report on the attachment program, which is evaluated at the departmental committee with pass or fail grade.

Course Code: ECEg5104

Course Name **Industrial Management and Engineering Economy**

Credit Hour: 3

Pre requisite Economics

Basics of management and functions of management; basics of plant layout, ergonomics and industrial safety; defining forecasting and the use of forecasting techniques; inventory management and control; basics of project management and resource allocation; project crashing and project risks; time value of money and cash flow; present and future worth of investments; basic concepts and computations of depreciation; cost comparison of alternative investments; accounting and budgeting fundamentals.

Course Code: ECEg518

Course Name **Final Year Project**

Credit Hour: 6

Pre requisite All senior standing courses

Success of the project is determined in large part by whether students have adequately solved the problem that they are working on.